IV SEMESTER	L	T	P	C
	-	-	3	1.5

20AM4L02 : MACHINE LEARNING APPLICATIONS LAB

COURSE OBJECTIVES:

- Make use of data sets in implementing the Machine Learning algorithms.
- Implement the machine learning concepts and algorithms in any suitable language of choice.
- Apply Machine Learning algorithms to real world problems.

COURSEOUTCOMES:

At the end of the course, the student will be able to,

CO1: Understand the implementation procedures for the Machine Learning algorithms.
CO2: Design python programs for various learning algorithms.
CO3: Apply appropriate data sets to the Machine Learning algorithms.
CO4: Identify and apply Machine Learning algorithms to solve real world problems.
CO5: Apply various clustering algorithms for various applications.
CO6: Analyze various Machine Learning algorithms and their applications.

LIST OF EXPERIMENTS:

1. Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples. Read the training data from a .CSV file.
2. Implement Simple Linear Regression
3. Implement Multivariate Linear Regression
4. Implement Logistic Regression
5. Implement Multivariate Logistic Regression
6. Data preprocessing for classification
7. Confusion matrix for a binary classifier.
8. Implement Support Vector Machines.
9. Write a program to implement the naïve Bayesian classifier for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets.
10. Write a program to implement k-Nearest Neighbour algorithm to classify the iris data set. Print both correct and wrong predictions.
11. Write a program to construct a Bayesian network considering medical data. Use this model to demonstrate the diagnosis of heart patients using standard Heart Disease Data Set.
12. Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.
13. Implement k-Mean's algorithm to cluster a set of data stored in a .CSV file.
14. Implement Random Forest models for automatic classification
15. Implement Ensemble Model to perform classification.

Text Books:

1. Andreas Muller, "Introduction to Machine Learning with Python: A Guide for Data Scientists", 1st Edition, O'Reilly, 2016.
2. Andriy Burkov, "The Hundred-Page Machine Learning Book", 1st Edition, Notion Press, 2019.

Reference Books:

1. Zhi-Hua Zhou, "Ensemble Methods: Foundations and Algorithms", CRC Press, 2012.
2. https://machinelearningmastery.com/stacking-ensemble-machine-learning-with-python/
3. https://sebastianraschka.com/blog/2018/model-evaluation-selection-part4.html